13 At the beginning of 2009, Mr Veale bought a company.
The value of the company was £50 000

Each year the value of the company increased by 2%.

(a) Calculate the value of the company at the beginning of 2017
Give your answer correct to the nearest £100

2017 – 2009 = 8 yearly increases. P = percentage.

	P	GBP	
	100	50000	
×1.02			×1.02
	102	51000	

Yearly multiplier = 102 ÷ 100 = 1.02

Value = 50000×1.02^8

1 mark
Use compound growth for 8 years

= £58582.969... → nearest £100: £58600

1 mark
Correct rounded value

£ 58600

(2)

At the beginning of 2009 the value of a different company was £250 000
In 6 years the value of this company increased to £325 000

This is equivalent to an increase of x% each year.

(b) Find the value of x.
Give your answer correct to 2 significant figures.

Six-year multiplier = 325000 ÷ 250000 = 1.3.

	P	GBP	
	100	250000	
×1.3			×1.3
	130	325000	

1 mark
Find the overall growth multiplier

Let m be the yearly multiplier: $m^6 = 1.3$.

$m = 1.3^{1/6} = 1.044697507...$

1 mark
Take the sixth root for one year

$x = (m - 1) \times 100 = 4.4697507...\%$

4.5

1 mark
Correct percentage to 2 significant figures

(3)

(Total for Question 13 is 5 marks)